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Using the Completing the Square Method

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Using the Completing the Squar...

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Using the Completing the Square Method

Some trinomials are perfect squares. When they are factored, the two binomial factors are identical. These trinomials are called **trinomial squares**.

$x^2 + 10x + 25 = (x + 5)^2$

Completing the square is a technique for solving quadratic equations that uses trinomial squares. Study the example to see how to make a trinomial square.

Example 1

Hide

Fill in the blank to make a perfect square.

$x^2 + 8x + \underline{\hspace{1cm}}$

Step 1

Take the number from the middle term: 8

Step 2

Divide by 2.

$8 \div 2 = 4$

Step 3

Square the result.

$4^2 = 16$

Step 4

Fill in the blank.

$x^2 + 8x + \textcolor{green}{16}$

Now you are ready to solve an equation by completing the square.

Example 2

Hide

Solve for x .

$x^2 - 12x + 20 = 7$

Step 1

Make a space to complete the square.

Subtract 20 from both sides of the equation.

$x^2 - 12x + 20 = 7$

$x^2 - 12x \underline{\hspace{1cm}} = -13$

Step 2

What number completes the perfect square?

Divide the number of the middle term by 2. Square the result.

$-12 \div 2 = -6$

$(-6)^2 = \textcolor{green}{36}$

Step 3

Fill in the blank. Add 36 to both sides.

$x^2 - 12x \underline{\hspace{1cm}} = -13$

$x^2 - 12x + 36 = 23$

Step 4

Factor the left side.

$(x - 6)^2 = 23$

Step 5

Take the square root of both sides.

$\sqrt{(x - 6)^2} = \sqrt{23}$

$x - 6 = \pm \sqrt{23}$

Step 6

Isolate the variable. Add 6 to both sides.

$x = 6 \pm \sqrt{23}$

The symbol $\pm$ means “plus or minus.”

There are two solutions: $6 + \sqrt{\textcolor{green}{23}}$ and $6 - \sqrt{\textcolor{green}{23}}$.

The symbol $\pm$ means “plus or minus.”

You can use your calculator to find the value of the solutions. Since the solutions represent irrational numbers, you will need to round to a convenient decimal place.

$6 + \sqrt{23} \approx 10.7958$

$6 - \sqrt{23} \approx 1.2042$

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